1	2	3	4	5	6	Total
/20	/20	/20	/20	/20	/20	/100

Name: ______________________ July 7, 2018. Saturday

Department/School:______________________ SID:______________________

University Physics

Final Examination

Kuang Yaming Honors School, Nanjing University

$$\Gamma(\nu)=\int_0^{+\infty} e^{-x}x^{\nu-1}dx,\quad \Gamma(\nu+1)=\nu\Gamma(\nu),\quad \Gamma(1)=1,\quad \Gamma(1/2)=\sqrt{\pi}.$$

<u>Select five out of following six problems.</u>

1.(20pts.) A rod of rest length L_0 moves with a speed v along the horizontal direction, i.e., the direction of the x-axis. The rod makes an angle of θ_0 with respect to the x'-axis in the S' frame in which the rod is at rest.

(a) Show that the length L of the rod as measured by a stationary observer is $L=L_0(1-\beta^2\cos^2\theta_0)^{1/2}$ where $\beta=v/c$.

(b) Show that the angle θ the rod makes with the x-axis as the stationary observer measured is determined by $\tan\theta=\gamma\tan\theta_0$ where $\gamma=(1-\beta^2)^{-1/2}$.

Solution: (a) In the S' frame : $L'_x=L_0\cos\theta_0$, $L'_y=L_0\sin\theta_0$. (1)

Then, as measured by the observer in S frame, we have

$$L_x=\sqrt{1-\beta^2}L'_x\,,L_y=L'_y. \quad (2)$$

Thus

$$L=\sqrt{L_x^2+L_y^2}=\sqrt{(1-\beta^2)L_x'^2+L_y'^2}=\sqrt{(1-\beta^2)L_0^2\cos^2\theta_0+L_0^2\sin^2\theta_0}=L_0(1-\beta^2\cos^2\theta_0)^{1/2}.$$

(b) From (2), we obtain

$$\tan\theta=\frac{L_y}{L_x}=\frac{L'_y}{\sqrt{1-\beta^2}L'_x}=\frac{L_0\sin\theta_0}{\sqrt{1-\beta^2}L_0\cos\theta_0}=\gamma\tan\theta_0.$$

2. (20pts.) A person on Earth observes two spacecrafts moving in the same direction toward the Earth. Spacecraft A has a speed of 0.50c, and spacecraft B has a speed of 0.80c.

(a) What is the speed v_A of spacecraft A measured by an observer in spacecraft B?

(b) What is the speed v_B of spacecraft B measured by the observer in spacecraft A?

Solution: (a) Taking the Earth as frame S and spacecraft B as frame S', we have $u=0.8c$ and $v_x=0.5c$. Thus

$$v_A=v'_x=\frac{v_x-u}{1-\frac{uv_x}{c^2}}=\frac{0.5c-0.8c}{1-\frac{0.8c\times0.5c}{c^2}}=-0.5c.$$

(b) Taking the Earth as frame S and spacecraft A as frame S', we have $u = 0.5c$ and $v_x = 0.8c$ .Thus

$$v_B = v_x' = \frac{v_x - u}{1 - \frac{uv_x}{c^2}} = \frac{0.8c - 0.5c}{1 - \frac{0.5c \times 0.8c}{c^2}} = 0.5c.$$

3.(20pts.) A block of ice of mass $1\,\mathrm{kg}$ and at temperature of $0^\circ\mathrm{C}$ is put in thermal contact with a grand heat bath at temperature of 25 °C. After some time, the ice and the heat bath are in equilibrium with each other. The latent heat of melting of ice (at $0^\circ\mathrm{C}$) is 3.35×10^5 J/kg and the specific heat capacity of water at constant pressure is 4.187×10^3 J/kg·K .

(a) Find ΔS_1, the change in entropy of the ice (water).

(b) Find ΔS_2, the change in entropy of the heat bath.

(c) Find ΔS_t, the change in entropy of the total system.

Solution: (a) $m = 1\,\mathrm{kg}$, $L = 3.35\times10^5$ J/kg, $c_p = 4.187\times10^3$ J/kg·K, $T_i = 273.15\,\mathrm{K}$, $T_f = 298.15\,\mathrm{K}$.

$$\Delta S_1 = \frac{mL}{T_i} + \int_{T_i}^{T_f} \frac{mc_p}{T} = \frac{mL}{T_i} + mc_p \ln\frac{T_f}{T_i} = 1.23\times10^3\ \mathrm{J/K} + 0.37\times10^3\,\mathrm{J/K} = 1.60\times10^3\ \mathrm{J/K}.$$

(b)

$$\Delta Q = -mL - mc_p\left(T_f - T_i\right) = -3.35\times10^5\ \mathrm{J} - 1.05\times10^5\ \mathrm{J} = -4.40\times10^5\ \mathrm{J}\ 1.05$$

$$\Delta S_2 = \frac{\Delta Q}{T_f} = -1.48\times10^3\ \mathrm{J/K}.$$

(c) $$\Delta S_t = \Delta S_1 + \Delta S_2 = 1.2\times10^2\ \mathrm{J/K}.$$

4. (20pts.) When a gas expands from volume V_0 to V at temperature T_0, the work done by the gas is given by

$$W = RT_0 \ln\frac{V}{V_0}$$

where R is the gas constant. And the entropy of the gas at temperature T and volume V is given by

$$S = R\frac{V_0}{V}\left(\frac{T}{T_0}\right)^{\alpha}$$

where α is a positive constant.

(a) Write down the differential form dF, where $F = F(T,V)$ is the Helmholtz free energy of the system.

(b) Taking (T_0,V_0) as the reference point and $F(T_0,V_0) = F_0$, find the Helmholtz free energy of the system at temperature T_0 and volume V, $F(T_0,V)$.

(c) Using the condition given in (b), find the Helmholtz free energy of the system at temperature T and volume V, $F(T,V)$.

(d) Find the internal energy U, and then the heat capacity at constant volume C_V of the system.

Solution: (a) The fundamental equation of thermodynamics reads

$$dU = TdS - pdV .$$

Making the Legendre transformation $F = U - TS$, we have

$$dF = d(U-TS) = TdS - pdV - TdS - SdT = -SdT - pdV . \qquad (1)$$

(b) From (1), we have that in the isothermal process at $T = T_0$

$$F(T_0,V) - F(T_0,V_0) = \int_{v_0}^{V} - pdV = -W = -RT_0 \ln\frac{V}{V_0} .$$

Thus

$$F(T_0,V) = F_0 - RT_0 \ln\frac{V}{V_0} \qquad (2)$$

(c) In the isochoric process $(T_0 V) \to (T,V)$, we have

$$F(T,V) - F(T_0,V) = \int_{T_0}^{T} - SdT = -R\frac{V_0}{V}\frac{1}{T_0^{\alpha}}\int_{T_0}^{T} T^{\alpha} dT = -\frac{R}{\alpha+1}\frac{V_0}{V}\frac{1}{T_0^{\alpha}}\left(T^{\alpha+1} - T_0^{\alpha+1}\right).$$

Thus

$$F(T,V) = F(T_0,V) - \frac{R}{\alpha+1}\frac{V_0}{V}\frac{1}{T_0^{\alpha}}\left(T^{\alpha+1} - T_0^{\alpha+1}\right) = F_0 - RT_0 \ln\frac{V}{V_0} - \frac{R}{\alpha+1}\frac{V_0}{V}\frac{1}{T_0^{\alpha}}\left(T^{\alpha+1} - T_0^{\alpha+1}\right)$$

(c)

$$U = F + TS = F_0 - RT_0 \ln\frac{V}{V_0} - \frac{R}{\alpha+1}\frac{V_0}{V}\frac{1}{T_0^{\alpha}}\left(T^{\alpha+1} - T_0^{\alpha+1}\right) + RT\frac{V_0}{V}\left(\frac{T}{T_0}\right)^{\alpha}$$

$$= F_0 - RT_0 \ln\frac{V}{V_0} + \frac{R}{\alpha+1}\frac{V_0 T_0}{V} + \frac{\alpha R}{\alpha+1}\frac{V_0}{V}\frac{T^{\alpha+1}}{T_0^{\alpha}} .$$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \alpha R\frac{V_0}{V}\left(\frac{T}{T_0}\right)^{\alpha}$$

$$\left(C_V = T\left(\frac{\partial S}{\partial T}\right)_V = \alpha R\frac{V_0}{V}\left(\frac{T}{T_0}\right)^{\alpha} \right)$$

5. (20 pts.) There is a system of gas with fixed number of molecules.

(a) Show that

$$dS = \frac{C_p}{T}dT - \left(\frac{\partial V}{\partial T}\right)_p dp$$

where S, T, p, V and C_p are the entropy, temperature, pressure, volume and heat capacity at constant pressure of the system, respectively.

(b) Show that $\left(\frac{\partial T}{\partial p}\right)_S = \frac{T}{C_p}\left(\frac{\partial V}{\partial T}\right)_p$.

(c) If the system is a monatomic ideal gas of n mole, find $(\partial T/\partial p)_S$. Will the temperature of the gas increase or decrease after an adiabatic expansion? Why?

(d) For a monatomic ideal gas of n mole, show that the state of system obeys the equation

$$Tp^{-\frac{2}{5}} = C$$

in the adiabatic process, where C is a constant.

Solution: (a)

$$S = S(T,p) \quad \Rightarrow \quad dS = \left(\frac{\partial S}{\partial T}\right)_p dT + \left(\frac{\partial S}{\partial p}\right)_T dp$$

From $dH = TdS + Vdp = T\left[\left(\frac{\partial S}{\partial T}\right)_p dT + \left(\frac{\partial S}{\partial p}\right)_T dp\right] + Vdp = T\left(\frac{\partial S}{\partial T}\right)_p dT + \left[T\left(\frac{\partial S}{\partial p}\right)_T + V\right]dp$,

we obtain $C_p \equiv \left(\frac{\partial H}{\partial T}\right)_p = T\left(\frac{\partial S}{\partial T}\right)_p$, or $\left(\frac{\partial S}{\partial T}\right)_p = \frac{C_p}{T}$.

From $G = U - TS + pV$, $dG = -SdT + Vdp$, we have the Maxwell relation $\left(\frac{\partial S}{\partial p}\right)_T = -\left(\frac{\partial V}{\partial T}\right)_p$.

Thus $dS = \frac{C_p}{T}dT - \left(\frac{\partial V}{\partial T}\right)_p dp$.

(b) Letting $dS = 0$, we have $\frac{C_p}{T}dT - \left(\frac{\partial V}{\partial T}\right)_p dp = 0$. Thus

$$\left(\frac{\partial T}{\partial p}\right)_S = \frac{T}{C_p}\left(\frac{\partial V}{\partial T}\right)_p .$$

(c) For n mole of ideal gas, $pV = nRT$ or $V = \frac{nRT}{p}$, thus we have

$$\left(\frac{\partial V}{\partial T}\right)_p = \frac{nR}{p}.$$

For n mole of monatomic gas $C_V = \frac{3}{2}nR$ and $C_p = C_V + R = \frac{5}{2}nR$, thus

$$\left(\frac{\partial T}{\partial p}\right)_S = \frac{T}{C_p}\left(\frac{\partial V}{\partial T}\right)_p = \frac{2}{5}\frac{T}{p}.$$

Since $\left(\frac{\partial T}{\partial p}\right)_S > 0$, the temperature of the gas will decrease after a reversible adiabatic expansion.

(c) In a reversible adiabatic process, $dS = 0$. Assuming that $T = T(S,p)$, we have

$$dT = \left(\frac{\partial T}{\partial S}\right)_p dS + \left(\frac{\partial T}{\partial p}\right)_s dp = 0 + \frac{2}{5}\frac{T}{p}dp.$$

Thus $\frac{dT}{T} = \frac{2}{5}\frac{dp}{p}$ or $\ln T = \ln p^{\frac{2}{5}} + \ln C$ or $Tp^{-2/5} = C$.

6. (20pts.) In an effusive beam of ideal gas, the number of molecules with z-component of velocity within $v_z \to v_z + dv_z$ that passes through the hole per unit area per unit time is given by

$$d\Gamma(v_z) = n_0\left(\frac{m}{2\pi k_B T}\right)^{1/2} v_z e^{-\frac{mv_z^2}{2k_BT}} dv_z, \quad v_z > 0$$

where n_0 the number density of the molecules in the container, m the mass of the molecule, T the temperature, and k_B the Boltzmann constant. Here the normal direction of the escaping hole is assigned as the direction of the z-axis.

(a) Calculate the average of the z-component of the velocity, $\langle v_z \rangle$, of the molecules in the effusive beam.

(b) Calculate $\langle v_z^2 \rangle$ and then the uncertainty (fluctuation) $\Delta v_z = \sqrt{\langle v_z^2 \rangle - \langle v_z \rangle^2}$.

(c) Show that the molecular flux of the effusive beam, i. e., the total number of molecules that passes through the hole per unit area per unit time is given by $\Gamma = p/\sqrt{2\pi m k_B T}$ where p is the pressure of the gas in the container.

Solution: (a)

$$\langle v_z \rangle = \frac{\int v_z d\Gamma}{\int d\Gamma} = \frac{\int_0^{+\infty} v_z^2 e^{-\frac{mv_z^2}{2k_BT}} dv_z}{\int_0^{+\infty} v_z e^{-\frac{mv_z^2}{2k_BT}} dv_z} \underset{\frac{mv_z^2}{2k_BT}=x^2}{=} \sqrt{\frac{2k_BT}{m}} \frac{\int_0^{+\infty} x^2 e^{-x^2} dx}{\int_0^{+\infty} x e^{-x^2} dx}$$

$$\underset{x^2=t}{=} \sqrt{\frac{2k_BT}{m}} \frac{\int_0^{+\infty} t e^{-t} d\left(\sqrt{t}\right)}{\int_0^{+\infty} \sqrt{t} e^{-t} d\left(\sqrt{t}\right)} = \sqrt{\frac{2k_BT}{m}} \frac{\int_0^{+\infty} \sqrt{t} e^{-t} dt}{\int_0^{+\infty} e^{-t} dt} = \sqrt{\frac{2k_BT}{m}} \frac{\Gamma\left(\frac{3}{2}\right)}{\Gamma(1)}$$

$$= \sqrt{\frac{2k_BT}{m}} \frac{1}{2}\sqrt{\pi} = \sqrt{\frac{\pi k_BT}{2m}}.$$

(b)

$$\langle v_z^2 \rangle = \frac{\int v_z^2 d\Gamma}{\int d\Gamma} = \frac{\int_0^{+\infty} v_z^3 e^{-\frac{mv_z^2}{2k_BT}} dv_z}{\int_0^{+\infty} v_z e^{-\frac{mv_z^2}{2k_BT}} dv_z} \underset{\frac{mv_z^2}{2k_BT}=x^2}{=} \frac{\int_0^{+\infty} x^3 e^{-x^2} dx}{\int_0^{+\infty} x e^{-x^2} dx}$$

$$\underset{x^2=t}{=} \frac{2k_BT}{m} \frac{\int_0^{+\infty} t^{3/2} e^{-t} d\left(\sqrt{t}\right)}{\int_0^{+\infty} \sqrt{t} e^{-t} d\left(\sqrt{t}\right)} = \frac{2k_BT}{m} \frac{\int_0^{+\infty} t e^{-t} dt}{\int_0^{+\infty} e^{-t} dt} = \frac{2k_BT}{m} \frac{\Gamma(2)}{\Gamma(1)}$$

$$= \frac{2k_BT}{m}.$$

$$\Delta v_z = \sqrt{\langle v_z^2 \rangle - \langle v_z \rangle^2} = \sqrt{\frac{2k_BT}{m} - \frac{\pi k_BT}{2m}} = \sqrt{\frac{(4-\pi)k_BT}{2m}}.$$

(c)

$$\Gamma = \int d\Gamma(v_z) = \int_0^{+\infty} n_0\left(\frac{m}{2\pi k_B T}\right)^{1/2} v_z e^{-\frac{mv_z^2}{2k_BT}} dv_z = n_0\left(\frac{m}{2\pi k_B T}\right)^{1/2} \int_0^{+\infty} e^{-\frac{mv_z^2}{2k_BT}} d\left(\frac{v_z^2}{2}\right)$$

$$= n_0\left(\frac{m}{2\pi k_B T}\right)^{1/2} \left(\frac{k_BT}{m}\right) \int_0^{+\infty} e^{-\frac{mv_z^2}{2k_BT}} d\left(\frac{mv_z^2}{2k_BT}\right) = \frac{n_0 k_B T}{\sqrt{2\pi m k_B T}} = \frac{p}{\sqrt{2\pi m k_B T}}.$$

($p = \frac{Nk_BT}{V} = n_0 k_B T$)